

Factor-Based Pareto Frontier Investing and Exchange Traded Funds

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Introduction

This paper looks to analyze Exchange Traded Funds (ETFs) through the lens of Fama and French's five factor asset pricing model. I will first define ETFs and describe the unique characteristics that appeal to investors. Then I will cover the history of asset pricing models culminating in the arrival at the five-factor asset pricing model as proposed by Fama and French in their 2014 paper. Using the knowledge of the five-factors, I will look to identify factor-exposed ETFs and test their recent historical performance against the S&P 500. Finally, I will propose a novel way to view and assess stocks using the background of the five factors in the form of a Pareto frontier analysis in hopes of discovering a way to empirically form diversified portfolios or ETFs. I will conclude by revealing limitations of the Pareto frontier analysis I performed and propose further avenues of interest and room for improvement and expansion of this analysis.

Exchange Traded Funds

Exchange Traded Funds (ETFs) are defined as pooled securities that can be bought and sold like a stock [1]. They are a common investment vehicle for those looking to invest in a diversified portfolio of assets. The most prevalent counterpart to ETFs is mutual funds which also offer diversified portfolios of assets, but there are some key differences.

The first difference is ETFs can be traded at any time during the trading day, like regular stock securities, whereas mutual funds can only be bought and sold at the end of the trading day. This also means that the price of ETFs is constantly being updated so purchasers have real time price information throughout the day, whereas mutual fund prices are only updated at the end of the trading day in time for the transactions to take place [3].

The other major difference is the exposure to asset classes. In a traditional mutual fund, you can typically only invest in stocks, bonds, and other securitized assets, whereas an ETF opens far more investment avenues. In addition to stocks and bonds, there are ETFs which allow investment in commodities, real estate, currency, and foreign markets. Furthermore, ETFs can also be combined with covered options strategies [3].¹

History of Asset Pricing

In 1965 Eugene Fama proposed the efficient market hypothesis (EMH) which states that share prices reflect all information available about a security and that consistently outperforming

¹ A covered option is a financial transaction in which the investor writes an option on an underlying security which they also own [4].

the market² is impossible without taking on excess risk [6]. Around the same time, the Capital Asset Pricing Model (CAPM) was also being developed. The CAPM seeks to explain asset returns in relation to market risk, assuming that the EMH is true.

For about 20 years the CAPM was the gold standard of asset pricing, but in 1985, Rosenberg, Reid, and Lanstein found that certain stocks have returns unexplained by exposure to the market risk outlined in the CAPM. This seemed to demonstrate that markets were not efficient.³ In 1993 Fama and French introduced the idea of a three-factor asset pricing model, aligned with the EMH, which considered the market risk factor as proposed in the CAPM and two new factors, size, and value⁴ [8]. Many more factors have since been researched and proposed.

The Five-Factor Asset Pricing Model

In 2014, building upon their work two decades earlier, Fama and French proposed a revised model, which added two new risk factors to the previous three, profitability, and investment. The risk factors identified and included in the five-factor asset pricing model are market, size, value, profitability, and investment.

The market risk factor refers to a company's exposure to broad market risk. All companies are exposed to market risk in some way. This factor is related to beta as described in the CAPM, where a company with greater exposure to market risk is expected to have a greater return [7].

The size risk factor refers to the size of a company as measured by its market capitalization (market cap). To test for the presence of the size factor, a portfolio is formed by going long on small stocks and short on big stocks, meaning that the return of the portfolio is the return of the small stocks *minus* the return of big stocks [7]. In Fama and French's research they found that diversified portfolios of this nature, on average, indicate that small stocks are expected to return higher [7]. The remaining factors will follow a similar structure.

The value risk factor refers to the value of a company, which Fama and French measured using the book-to-market ratio.⁵ A portfolio is formed by going long on high B/M stocks and short on low B/M stocks. Fama and French found that high B/M stocks are expected to return higher [7].

The profitability risk factor refers to the operating profit of a company. A portfolio is formed by going long on high profitability stocks and short on low profitability stocks.

² Otherwise known as "generating alpha." Alpha refers to excess returns over the market return achieved without taking on excess risk.

³ i.e., this finding seemed to demonstrate that the efficient market hypothesis was false.

⁴ Risk factor terms will be defined in the next section, "The Five-Factor Asset Price Model."

⁵ The book-to-market ratio...

Unsurprisingly, Fama and French found that high profitability stocks are expected to return higher [7].

Finally, the investment risk factor refers to the total liabilities of a company. A portfolio is formed by going long on low investment stocks and short on high investment stocks. Fama and French found that low investment stocks are expected to return higher [7].

Factor-Exposed ETFs

With the knowledge of risk factors, institutions have begun offering unique ETFs composed of risk factor-exposed securities which seek to outperform the market over the long term. Some examples include [9][10]:

- AVUV: Avantis US Small-Cap Value ETF
- AVDV: Avantis International Small-Cap Value ETF
- VBR: Vanguard Small-Cap Value Index Fund ETF
- VBK: Vanguard Small-Cap Growth Index Fund ETF
- VTV: Vanguard Value Index Fund ETF

Many factor-exposed ETFs are quite new, having only existed for the past few years, meaning that the sample size for whether we should expect them to outperform over the long term is small. Given the findings of Fama and French, however, we would expect that ETFs like those listed would outperform the market over the long term. Luckily, the Vanguard ETFs have existed for at least ten years and therefore we can analyze whether, over this medium-term period, they have outperformed.

Vanguard Examples

For this analysis, we will consider the three Vanguard ETFs listed in the previous section, VBR, VBK, and VTV, as well as one more, VUG.⁶ The composition of each of these portfolios is as follows:⁷

⁶ VUG: Vanguard Growth Index Fund ETF. Both VTV and VUG are comprised of mostly large-cap stocks and thus can be considered large-cap value and growth ETFs respectively, counterparts to VBR and VBK.

⁷ Note that the stock composition of the Vanguard ETFs is not entirely disjoint, but this is not a problem because the overlap is small, and they still serve as useful examples of what employing factor-based investment strategies would look like.

Chart 1. Composition and measurement metrics for the four Vanguard factor-exposed ETFs, VBR, VBK, VTV, and VUG.

Small-cap Value

Fundamental	VBR
Number of stocks	855
Median market cap	\$6.4 B
P/E ratio	13.7x
P/B ratio	1.8x

Small-cap Growth

Fundamental	VBK
Number of stocks	637
Median market cap	\$8.0 B
P/E ratio	29.0x
P/B ratio	4.1x

Large-cap Value

Fundamental	VTV
Number of stocks	350
Median market cap	\$119.0 B
P/E ratio	18.4x
P/B ratio	2.7x

Large-cap Growth

Fundamental	VUG
Number of stocks	208
Median market cap	\$715.5 B
P/E ratio	37.3x
P/B ratio	10.3x

As these ETFs represent index funds of stocks with certain factor compositions, the number of stocks in each is very large. The other noteworthy item when looking at the composition and metrics of these portfolios is the use of P/B ratio to segregate value and growth stocks. This ratio is related to the B/M ratio that Fama and French used in their analysis, however the numerator and denominator are effectively flipped meaning that in the Vanguard ETFs, low P/B ratio constitutes value stocks and conversely high P/B ratio constitutes growth stocks.

To analyze the performance, we will first calculate the average yearly log-return of each ETF from weekly historical pricing data using the formula:

$$r_t = \ln\left(\frac{P_{t+1}}{P_t}\right)$$

Where P_t is the price of the ETF at time t . The table of average yearly log-returns is as follows:

Table 1. Average yearly log-returns for the four factor-exposed Vanguard ETFs.

Return	Value	Growth
Small	8.43%	7.57%
Large	9.31%	13.13%

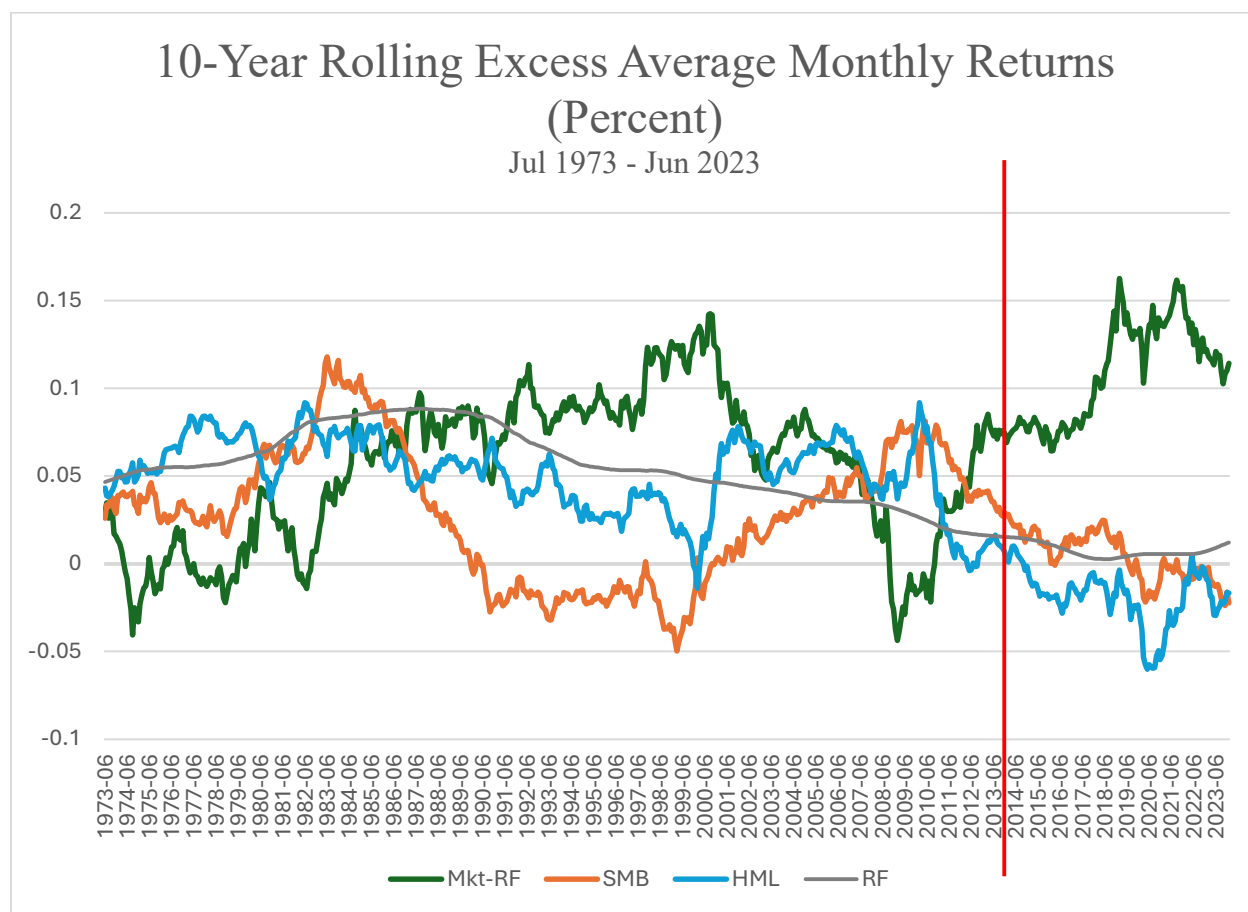
In this table we can see that the VBR small-cap value ETF had the second lowest average yearly log-return over the past ten years, but more importantly, it had a much lower return than VUG large-cap growth ETF which returned by far the highest over the past ten years. Given the

previous findings of Fama and French, we would expect small-cap value stocks to have the highest return and large-cap growth stocks to have the lowest return [7]. The reason I posit for this is the recent overperformance of the market risk factor combined with recent negative excess returns for the size and value risk factors.⁸

Market Factor Overperformance and Negative Excess Returns of Size and Value Factors

As stated in the previous section, the market risk factor has returned exceptionally well over the past ten years and the size and value factors have experienced negative excess returns. Both phenomena are demonstrated in the following graph:⁹

Chart 2. 10-year rolling average monthly excess returns in percent for the market, size, and value factors, including the 10-year rolling average risk-free rate.



As evidenced in the above graph, over the past ten years the excess return of the market factor has risen to record highs, the size factor has been depressed to around zero excess return, and the value factor has fallen to a record low level of excess return. The reason for this is the

⁸ Size and value risk factors yielding negative excess returns means that, in the case of size, large stocks performed better than small stocks, and in the case of value, growth stocks performed better than value stocks.

⁹ **Chart A1** in the appendix is a comprehensive graph showing excess returns of all five risk factors. This graph shows the three risk factors discussed in the Vanguard example for simplicity and readability.

generally bullish sentiment from investors, dating back as far as the end of the great recession. Since 2010, the market has experienced one of the longest running bull markets in history, one where we have seen investors favor large, growth technology companies, driving the excess return of the market risk factor higher and depressing the size and value risk factors. This graph explains the previously unexpected results from the Vanguard ETFs analysis. These conditions proved to be a substantial driver for the exceptional returns of VUG, which consists of the large-cap growth companies that have experienced a boom over the past ten years.

There is another important note to be made about this graph. Looking over the long term, we can see that the market factor has not always performed above the size and value factors. Furthermore, at no point over the past fifty years have all three factors experienced negative excess returns at the same time.¹⁰ Therefore, an investor with a long-term time horizon may consider diversifying a portfolio by investing in factor-exposed securities.

Sharpe Ratios

Another useful metric to analyze the performance of the Vanguard ETFs is the Sharpe ratio. The Sharpe ratio is a common metric which investors use to analyze the return versus the volatility in a portfolio. The formula for the Sharpe ratio is:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Where R_p is the return on portfolio p , R_f is the average annual return of ten-year treasuries, and σ_p is the volatility of portfolio p . Calculating the average yearly Sharpe ratio for each Vanguard ETF fills the following table:¹¹

Table 2. Average yearly Sharpe ratio for the four factor-exposed Vanguard ETFs.

Sharpe Ratio	Value	Growth
Small	0.31	0.25
Large	0.48	0.58

For comparison's sake, the S&P 500 average yearly Sharpe ratio to be 0.59, right on par with VUG's 0.58 Sharpe ratio. Once again, the relatively low Sharpe ratio of the VBR compared to the S&P 500, VUG, and even VTV, illustrates the bullish market sentiment driving the excess

¹⁰ By extension, at no point over the past fifty years have all five factors experienced negative excess returns at the same time (see **Chart A1**).

¹¹ A higher Sharpe ratio is better, because it means that the portfolio is generating a high return relative to its volatility. In the Sharpe ratio, volatility serves as an imperfect proxy for portfolio risk.

return of the market factor higher and the negative excess return experienced by the size and value factors.

Pareto Frontier Factor-Based Analysis

The previous sections serve as the motivation for the analysis I am going to perform next. I find that the returns and Sharpe ratios of the Vanguard factor-exposed index-style ETFs unsatisfactory. I am going to explore a different way of identifying and selecting multi-factor exposed stocks for the formation of a hypothetical ETF. The method used is called a Pareto frontier analysis and serves as an alternative way to test the validity of Fama and French's five-factor model.

The idea of the following Pareto analysis is to see whether selecting stocks based on objective factor-related metrics has any validity. This methodology uses historical metrics to determine where to set optimal cutoff points going forward. While past performance isn't an indicator of future return, this analysis seeks to investigate whether there are intrinsic numeric properties that high performing stocks share.

The general steps in the Pareto frontier process are:

1. Pick a *parameter* which will be used for iteration.
2. Pick a *success identifier*.
3. Pick at least one *diversity metric* which needs to be balanced.
4. Pick at least two *measurement values* that will be used to assess the tradeoff between the success identifier and diversity metric.
5. Iterate through the parameter values, calculating both measurement values at each iteration and create a scatter plot.
6. Identify the points which make up the Pareto frontier.
7. Select the point and accompanying *cutoff value* best suited to your needs.

These steps, just in theory, are nebulous; this process will become clear in the following example.

Pareto Frontier Example

Consider a scenario where we wish to create a hypothetical ETF of stocks based on the size and value factors. For this example, a subset of 20 randomly chosen NYSE stocks will be used.¹² The market-cap will serve as the parameter, the P/B ratio¹³ will define the diversity metric, and the stocks will be split along the median of the P/B ratio to segregate growth and value groups. A binary outperformance variable will indicate whether a stock outperformed the

¹² The one requirement of the randomly chosen stocks is that they have historical data dating back to at least January 2014.

¹³ A stock's market-cap and P/B ratio are taken from the closest date with available data to January 1, 2020.

S&P 500 over the period between January 2015 and December 2019.¹⁴ In this example, the 20 stocks will be sorted in descending order by market cap. We will then iterate through each portfolio composition in the following manner such that:

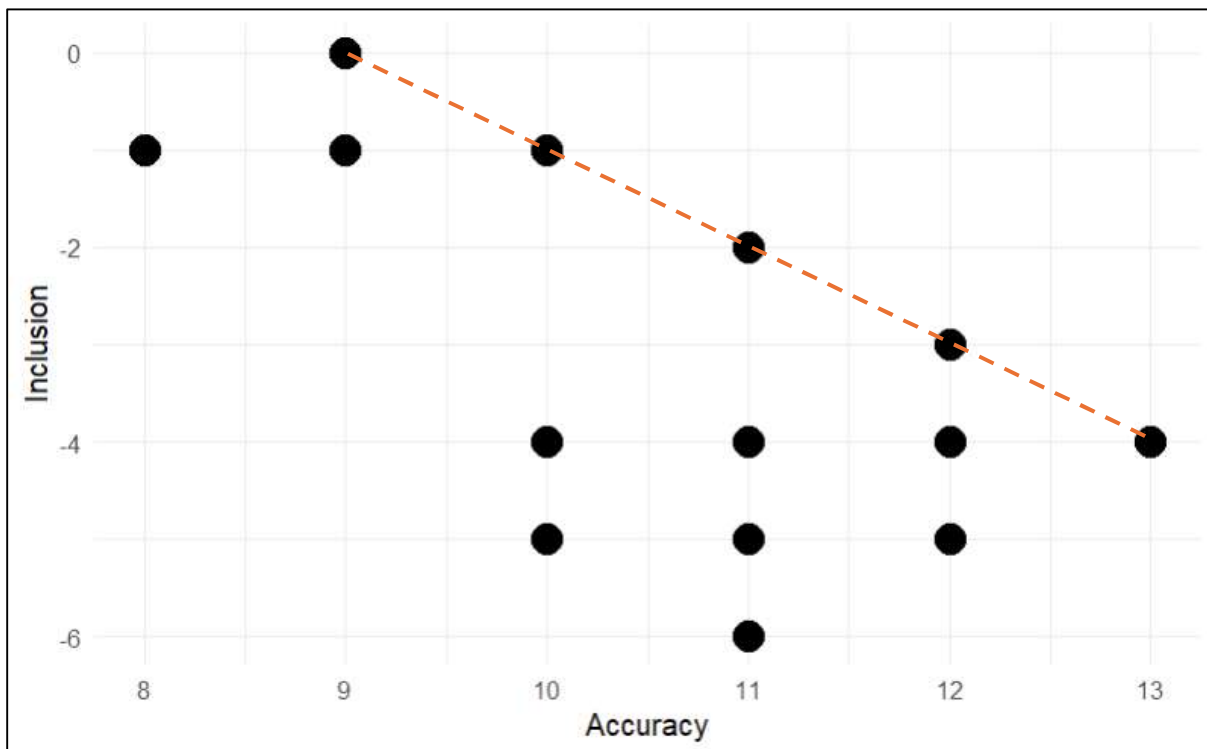
1. The first portfolio contains only the first largest stock by market cap.
2. The second portfolio contains the two largest stocks by market-cap.
3. The third portfolio contains the three largest, and so on.
4. The final, 20th portfolio will contain all 20 stocks.

At each iteration, we will calculate two portfolio measurement values, *accuracy*, and *inclusion* defined as follows:

- *Accuracy*: The sum of the number of correctly selected outperforming stocks plus correctly rejected underperforming stocks.
- *Inclusion*: The difference of the number of correctly included value stocks minus correctly included growth stocks.

Each portfolio's measurement values will then be plotted on an x-y plane, with the x-axis measuring accuracy and the y-axis measuring inclusion. Below is the plot for this example:¹⁵

Chart 3. Scatter plot of portfolios by measurement metrics *accuracy* vs *inclusion*.



¹⁴ 1 = outperformed; 0 = underperformed.

¹⁵ Note that there are only 14 points plotted for 20 portfolios. This is because it is possible for multiple portfolios to have the same *accuracy* and *inclusion* values.

On the plot, the dashed orange line represents the Pareto frontier. Along this curve are the Pareto optimal points for which you cannot improve one measurement value without worsening the other.¹⁶ All points which do not fall along the curve are considered suboptimal because you can improve one, or both measurement values without penalty. The curve is up and to the right because we are aiming to maximize for both accuracy and inclusion.¹⁷ The next step is to identify each Pareto optimal point's market-cap parameter cutoff value. This can be done by looking at the following table:

Table 3. Stock ticker symbol, market cap as of available data closest to January 1, 2020, value type, average yearly log-return, whether it outperformed the S&P 500, and the *accuracy* and *inclusion* values for each portfolio containing stocks up to and including the one on the line of the measurement.

ticker	mkt_cap_2020	value	average_log_return	outperform	Accuracy	Inclusion
GOOGL	934640	Growth	0.185463984	1	9	-1
USB	88820	Value	0.080878560	0	8	-1
BX	37870	Value	0.166582438	1	9	0
LULU	30330	Growth	0.285196548	1	10	-1
WY	21980	Value	0.007467279	0	9	-1
BKR	16740	Value	-0.064919448	0	8	-1
MGM	16490	Value	0.098341055	1	9	0
BR	14200	Growth	0.215496260	1	10	-1
LDOS	14130	Growth	0.249854200	1	11	-2
ZBRA	13770	Growth	0.239153916	1	12	-3
GNRC	6320	Growth	0.153448512	1	13	-4
JWN	6190	Growth	-0.087784354	0	12	-4
MSM	4310	Value	0.019618133	0	11	-4
FHI	3170	Value	0.041553041	0	10	-4
CROX	2860	Growth	0.242408533	1	11	-5
ALGT	2770	Growth	0.047294157	0	10	-5
AAON	2620	Growth	0.166577196	1	11	-6
GTLS	2410	Value	0.136166885	1	12	-5
CFFN	1880	Value	0.081702268	0	11	-5
RUSHB	1660	Value	0.100573824	1	12	-4

Two interesting phenomena are observed in this example. One, the Pareto optimal points (9,0) and (10,-1) appear twice, and two, there is a sequence where all five pareto optimal points come one after the other when iterating through market-cap. These phenomena are specific to this example and will rarely occur. Comparing the cutoff values selected at point (9,0) and

¹⁶ E.g., at the point (11,-2) if we wish to improve accuracy to 12, we must worsen inclusion to -3.

¹⁷ Maximizing inclusion in this example is to include more outperforming value stocks than outperforming growth stocks.

(13,-4) reveals two of the potential market-cap cutoff values for determining which stocks to include, 16.49 billion and 6.32 billion.

Testing Pareto ETF Performance

Using the portfolio composed with a cutoff value of 6.32 billion, let's test its performance on out of sample data. In other words, how does this portfolio of 11 stocks perform from January 2020 to April 2024? The average yearly log-return and volatility versus the S&P 500 are displayed in the table below:

Table 4. Average yearly log-return and volatility of the hypothetical Pareto ETF vs the S&P 500.

	Pareto ETF	S&P 500
Average Yearly Log-Return	8.30%	11.0%
Average Yearly Volatility	41.80%	22.81%

Over this 4.5-year timespan the Pareto ETF performed decently yielding an average annual log-return of 8.3% but still underperforming the S&P 500. More alarmingly, however, is the Pareto ETF's high volatility, almost twice that of the S&P 500. The primary reason for this high volatility is the small number of stocks included in the portfolio, a mere 11. It is well known that volatility only begins to settle when around 30 stocks are included in a portfolio [15]. This will be fleshed out further in the section "Limitations."

Testing Rejected Sample Performance

To test whether the model is able to select stocks any better than randomly guessing, I will look at the performance of a portfolio comprised of the rejected stocks.

Table 5. Average yearly log-return and volatility of the hypothetical Pareto ETF and a portfolio of stocks rejected by the model.

	Pareto ETF	Rejected Portfolio
Average Yearly Log-Return	8.30%	5.35%
Average Yearly Volatility	41.80%	49.79%

An unpaired two-sample t-test on the difference in average yearly log-return reveals a p-value which is not significant at a 90% confidence level and easily rejects this model.

However, this may be due to the small sample size of each portfolio, which will also be addressed in the following section.

Limitations

The primary limitation in this example analysis was the lack of readily available historical metrics. The historical market-cap data was unable to be scraped and thus needed to be sourced manually. The data that could be scraped, historical P/B ratios, required code which took thirty seconds per stock to scrape. Both of these limitations are the reason for the use of the small sample size of just 20 stocks in this example.

Furthermore, in this example the market-cap and P/B ratio are dynamic and change throughout time, therefore it might not be best practice to simply take the values from the final date of the historical period. This limitation can be overcome by periodically reclassifying stocks and rebalancing portfolios, say on a yearly basis, but this was beyond the scope of this analysis.

One of the potential limitations of using a Pareto analysis specifically on factors outlined in Fama and French's paper is that the factors are generated from a purely statistical analysis of stock returns. There is no solid explanation for *why* a stock's exposure to certain risk factors should yield excess return, just that there is some statistical evidence explaining they do. This is not a major problem in general for a Pareto frontier analysis because such analyses have broad applicability to any metric of company performance. Thus, we are not necessarily pigeonholed into the factors that Fama and French identified as important.

Finally, this example analysis uses a simple Pareto frontier to select stocks. Therefore, while nicely demonstrating the process of one of these analyses, it does not yield extraordinarily meaningful results. Regardless of these limitations, there is nonetheless ample room to improve upon and expand this idea.

Expanding This Idea

The very first expansion, before considering any others, is to increase the sample size. With a larger number of stocks, more concrete findings are expected to emerge. We will be able to see whether the volatility of a selected portfolio can be decreased. We can also retest whether a statistically significant difference between average yearly log-return of a Pareto portfolio and a rejected portfolio emerges. For the rest of the expansions mentioned below, we will assume the sample size is large.¹⁸

Here we only looked at market-cap and P/B ratio, which was split on the sample data median. We could instead split the P/B ratio according to the median of all NYSE stocks. Additionally, the generalizability of the Pareto frontier can be applied to almost any imaginable stock or company metric. Selecting and performing Pareto analyses on a variety of metrics

¹⁸ More precisely, imagine the expanded ideas being performed on all NYSE stocks.

would reveal stocks that perform well across many different metrics and portfolios of such stocks can subsequently be aggregated.

Furthermore, a Pareto frontier will also emerge in higher dimensional space, meaning we can take a plethora of different stock metrics and analyze them in a single model creating a higher dimensional Pareto frontier of optimal portfolios for us to choose from. This is a similar methodology to the aggregation method, but rather than analyzing each metric individually and aggregating the best stocks, we analyze the metrics all at once, which may generate a different, potentially better, set of Pareto optimal portfolios.

We also can alter the accuracy and inclusion metrics. For example, rather than using simple addition- and subtraction-based definitions we can use ratio-based measurements.¹⁹ We also have the option to ditch the accuracy and inclusion measurement values entirely and opt for different methods of measuring portfolio optimization.

Finally, we can change the binary identifier to be non-binary, which would also cause a movement into higher dimensions. In our example we considered overperformance and underperformance, but we could have instead considered overperformance, positive-return underperformance, and negative return. We can also change what the identifier compares the stock to altogether.

These are just a handful of ways in which the idea of performing a Pareto frontier analysis on stock metrics can be expanded, and there are surely others which I have not yet considered.

Conclusion

ETFs are a low-cost way for investors to create a diversified portfolio. The plethora of ETF offerings allow investors to invest their money according to each of their respective preferences. Some of the more recent and unique ETF offerings are composed according to Fama and French's risk factors.

While factor-exposed ETFs exist, several are extremely new to the scene having just been created within the last few years. The lack of significant historical data means that these ETFs are not yet proven to outperform the market over the long term, or at the very least, are not yet proven to have long-term average yearly Sharpe ratios better than the S&P 500. For the Vanguard ETFs, which have been around for longer and thus have some history to analyze, it seems that the size-value composed ETF has not performed as would be expected based on Fama and French's analysis, but as was shown this is due to historically low excess returns of the size and value factors. It remains to be seen whether these ETFs can perform over even longer time horizons.

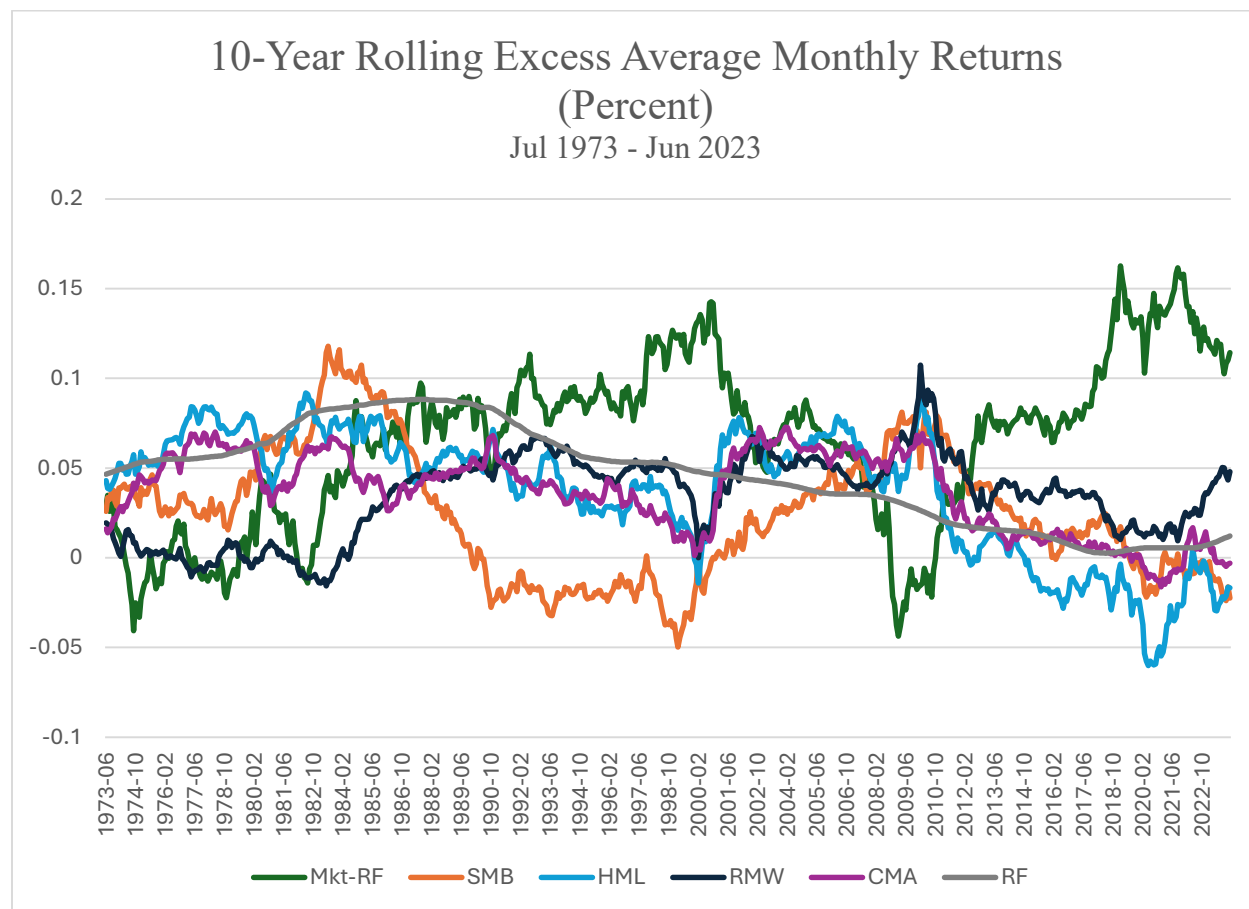
¹⁹ E.g., we could look at the percentage of growth and value stocks comprising each portfolio rather than the difference.

A different approach to forming portfolios or ETFs based on factors is with a Pareto frontier analysis. A simplistic analysis yielded inconclusive results and barring time constraints it would have been possible to perform more experiments with more data and flesh out whether there is merit to using a Pareto frontier analysis to select stocks for a portfolio. This experiment likely just scratches the surface with what's possible using a Pareto frontier analysis on stocks and certainly did not do it justice.

It is wise for investors to approach ETFs, factor-based investing, and Pareto frontier analyses with a curious yet critical eye, recognizing both the potential benefits and inherent uncertainties of these portfolio construction methods.

Appendix

Chart A1. 10-year rolling excess return in percent of all five factors (market, size, value, profitability, and investment), and the 10-year rolling average risk-free rate.



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